EDM Task 2

Fairness in Criminal Justice

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1. COMPAS Evaluation Framework

The modern criminal justice system faces the challenge of making complex decisions under uncertainty, such as assessing the risk of recidivism of a defendant. In this context, machine learning-based tools have emerged, such as the COMPAS system, whose goal is to estimate the probability that an individual will reoffend.

The use of these systems has generated an intense debate. On the one hand, supporters argue that they can improve efficiency and reduce human bias in judicial decision-making. On the other hand, there are concerns about their fairness, since these models are trained on historical data that may reflect pre-existing social and racial inequalities.

This paper analyzes the fairness of the COMPAS system using real data collected in Broward County (Florida) and published by ProPublica. The objective is to evaluate whether the model's predictions are fair across different racial groups, focusing especially on the concepts of sufficiency (calibration) and separation (equality of error rates).

To do so, various R programming libraries will be used, including tools for data manipulation, visualization, and fairness analysis. In addition, specific metrics will be applied to compare the model's behavior between African American and Caucasian individuals. This analysis will provide a better understanding of the limitations of predictive systems in sensitive contexts and encourage reflection on their responsible use in decision-making within the judicial system.

2. COMPAS Fairness and Risk Analysis

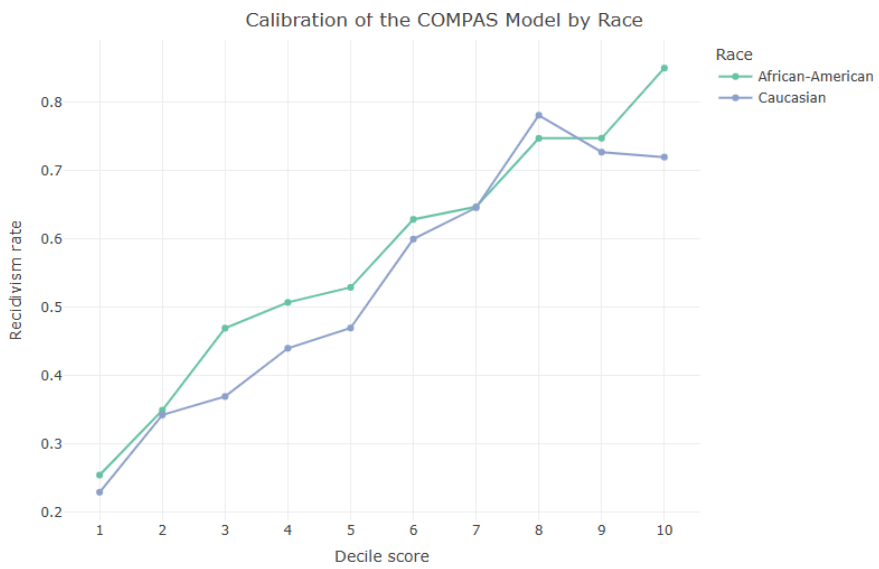
2.1. COMPAS Fairness Evaluation

The sufficiency property of the COMPAS model is evaluated by analyzing whether a given risk score corresponds to the same actual probability of recidivism across different racial groups. This involves examining the relationship between the risk score (decile_score) and observed recidivism (is_recid) for African-American and Caucasian individuals.

The criterion is assessed by verifying the following condition:
P(Y = 1 | R = r, A = a) = P(Y = 1 | R = r, A = b)

In this context, it can be expressed as:
P(is_recid = 1 | decile_score = r, race = African-American) = P(is_recid = 1 | decile_score = r, race = Caucasian)

The objective is to determine, for a fixed value of decile_score, whether the proportion of individuals who reoffend is the same across both racial groups, rather than comparing overall recidivism rates.



The results show that recidivism rates increase as the assigned risk score rises for both groups, and the trends are broadly similar, indicating a consistent behavior of the model. However, the curves are not identical, revealing differences at certain risk levels and suggesting imperfect calibration.

For example, at decile_score = 4:
44% of Caucasian individuals reoffend within 2 years.
50% of African-American individuals reoffend within 2 years.

These differences indicate that the same score does not always correspond to the same empirical probability of recidivism across groups.

Overall, the COMPAS model approximately satisfies the sufficiency property, as the general trends are aligned. However, the presence of numerical discrepancies shows that calibration is not exact, and therefore fairness under this criterion is only partially achieved.

The separation property is evaluated by examining whether the COMPAS model produces similar error rates across racial groups. This requires that individuals with the same true outcome receive similar predictions regardless of group membership.

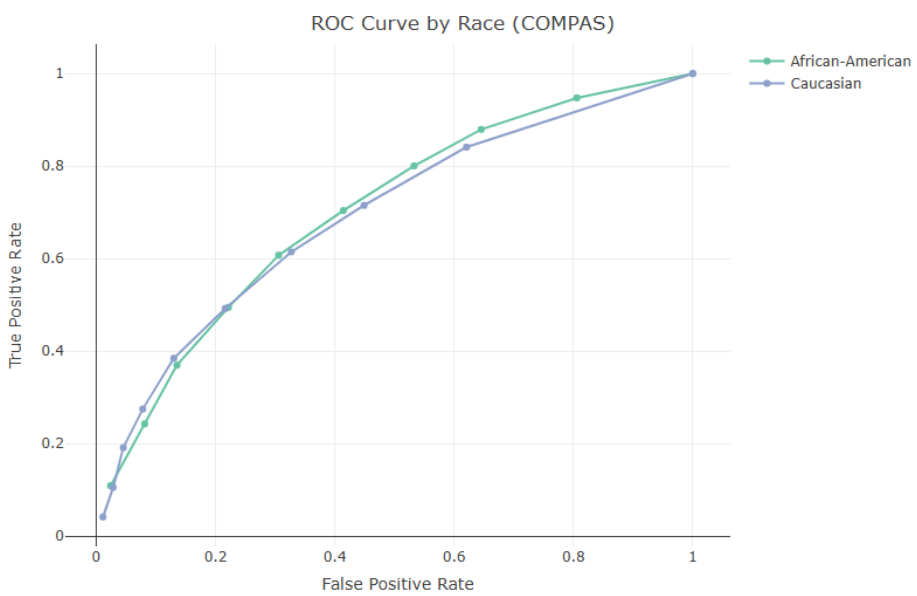
Formally, separation requires:
P(R = 1 | Y = y, A = a) = P(R = 1 | Y = y, A = b)
P(R = 0 | Y = y, A = a) = P(R = 0 | Y = y, A = b)

In this context, the analysis focuses on comparing:

True Positive Rate (TPR):
P(decile_score = 1 | is_recid = 1, race = African-American) = P(decile_score = 1 | is_recid = 1, race = Caucasian)

False Positive Rate (FPR):
P(decile_score = 1 | is_recid = 0, race = African-American) = P(decile_score = 1 | is_recid = 0, race = Caucasian)

The objective is to determine whether the model achieves equal predictive performance across groups when conditioning on the true outcome. This comparison is visualized using a ROC curve, where TPR and FPR for both groups are plotted.



The ROC curves for African-American and Caucasian individuals are very close across most thresholds, indicating similar predictive behavior. However, proximity of the curves does not guarantee that separation holds under a single common threshold, as equality of error rates must be satisfied for a specific decision rule.

To satisfy separation, the following conditions must hold with a difference below 1%:
TPR African-American = TPR Caucasian
FPR African-American = FPR Caucasian

The results identify three threshold pairs that approximately meet these conditions:
($t_{aa} = 1$, $t_{ca} = 1$) (trivial case)
($t_{aa} = 10$, $t_{ca} = 9$)
($t_{aa} = 7$, $t_{ca} = 5$)

Excluding the trivial case, these combinations show that similar error rates can be achieved by applying different thresholds to each group.

Using ($t_{aa} = 7$, $t_{ca} = 5$), the Positive Predictive Value (PPV) is analyzed:
$PPV = P(Y = 1 \mid R = 1)$

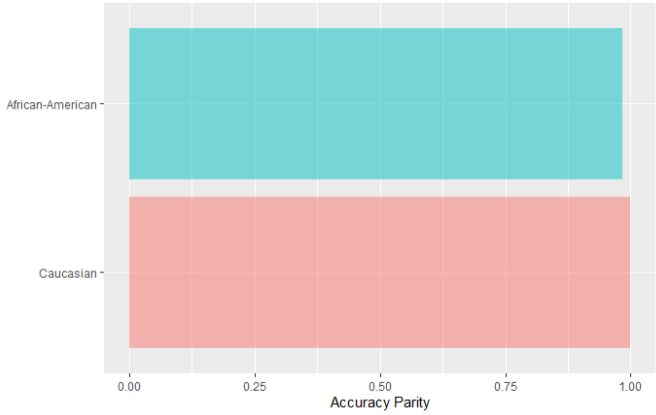
PPV measures the proportion of individuals classified as high risk who actually reoffend. The results show that PPV differs across groups even when TPR and FPR are approximately equalized.

This indicates that separation is not satisfied under a single common threshold. Although it can be approximated by adjusting thresholds for each group, this leads to disparities in PPV, meaning that a “high risk” prediction does not carry the same interpretation across groups.

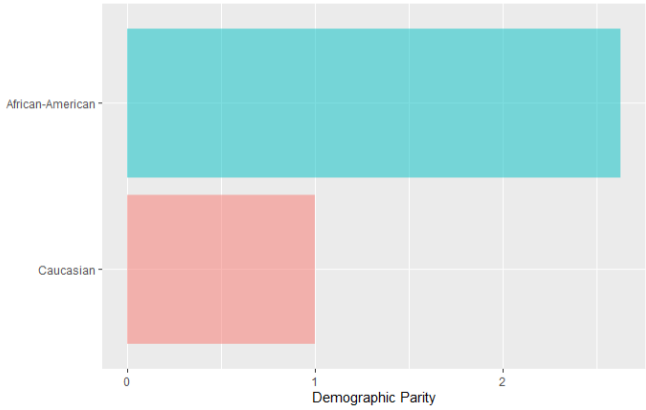
This outcome highlights a fundamental limitation of fairness in predictive systems: different fairness criteria cannot be simultaneously satisfied. Achieving approximate separation introduces inconsistencies in sufficiency, illustrating the inherent trade-off between these definitions of fairness.

2.2. Multi-Metric Fairness Analysis of COMPAS

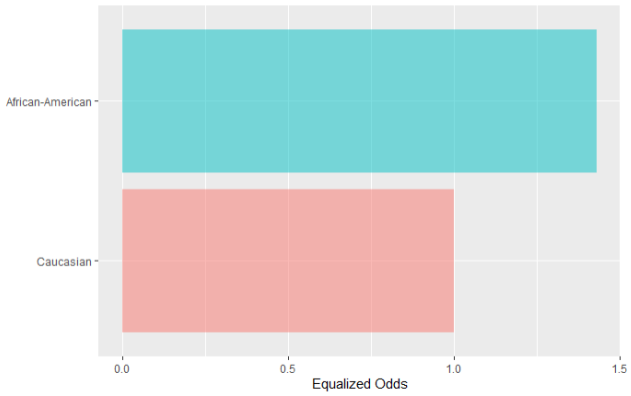
The R fairness package is used to evaluate the COMPAS model across African-American and Caucasian groups under several fairness definitions, since no single metric is sufficient to determine overall fairness.



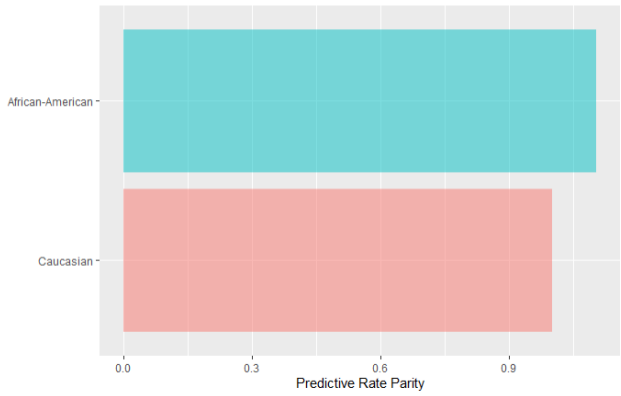
Accuracy parity compares overall classification accuracy between groups, defined as the proportion of correct predictions (true positives and true negatives). The Caucasian group is used as the reference baseline (value = 1), while the African-American group shows a slightly lower value. This indicates marginally lower overall accuracy for this group. However, accuracy parity alone is not sufficient to assess fairness, as it aggregates different types of errors without distinguishing between false positives and false negatives.



Demographic parity evaluates whether the probability of receiving a positive prediction is independent of race. The Caucasian group serves as the baseline (1), while African-American individuals receive approximately 2.5 times more “high risk” predictions. Therefore, $P(R = 1 \mid A = \text{African-American}) \approx 2.5 \times P(R = 1 \mid A = \text{Caucasian})$, indicating that demographic parity is not satisfied. However, this measure does not account for whether predictions are correct.



Equalized odds evaluates whether TPR and FPR are equal across groups. The results show differences in true positive rates between African-American and Caucasian individuals, confirming that separation is not satisfied. The African-American group exhibits a higher TPR relative to the baseline (Caucasian = 1), meaning a higher proportion of actual recidivists are correctly identified. This is consistent with previous findings that TPR and FPR differ across groups for the same threshold.



Predictive rate parity assesses whether the Positive Predictive Value (PPV), defined as P(Y = 1 | R = 1), is similar across groups. The results show that PPV differs between groups, indicating that sufficiency is not fully satisfied. The African-American group presents a slightly higher value than the baseline (Caucasian = 1), meaning that among individuals classified as high risk, a higher proportion actually reoffends. This aligns with previous results showing that the model is reasonably calibrated, but not perfectly.

The analysis using the fairness package shows that the COMPAS model does not simultaneously satisfy the main definitions of fairness:

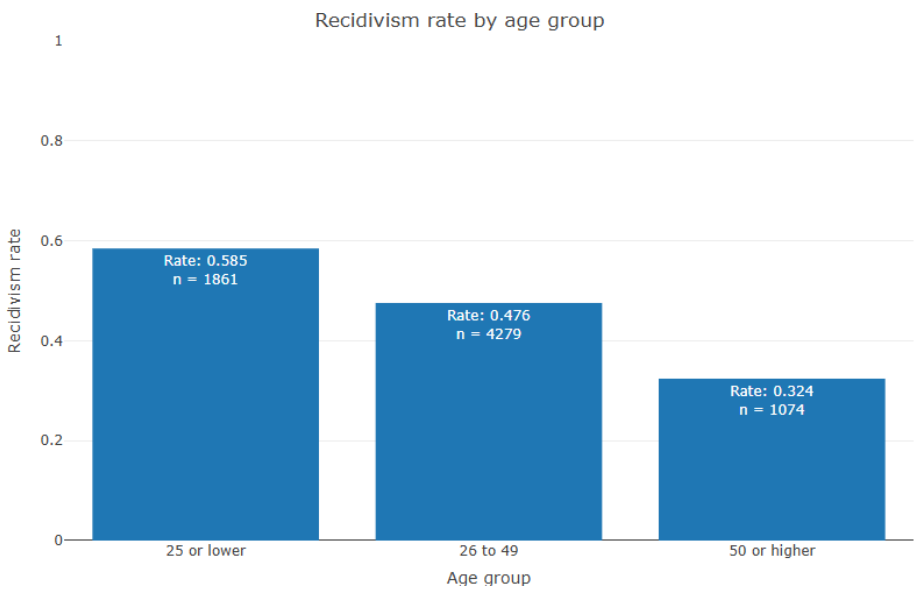
- It does not satisfy independence (demographic parity)
- It does not satisfy separation (equalized odds)
- It does not satisfy sufficiency (predictive rate parity)
- It shows differences in accuracy between groups (accuracy parity)

This confirms a key result in fairness theory: it is not possible to optimize all fairness metrics at the same time, as they are mutually incompatible in real-world settings. Therefore, any decision about which definition to prioritize depends on the ethical framework and the application context of the model, especially in highly sensitive domains such as criminal justice.

2.3. Age as a Risk Factor in Recidivism Analysis

The relationship between age and the probability of recidivism is examined to identify whether age can be considered a relevant risk factor. The analysis compares two groups: individuals aged 25 or younger and individuals aged 50 or older.

The objective is to compute and contrast the recidivism rates of both groups in order to detect significant differences associated with age and to assess its potential relevance in risk prediction within the criminal justice system.



The results show a clear disparity between the two groups. Individuals aged 25 or younger exhibit a significantly higher recidivism rate than those aged 50 or older, with a difference of approximately 25 percentage points. This suggests that age is strongly associated with recidivism risk in the dataset.

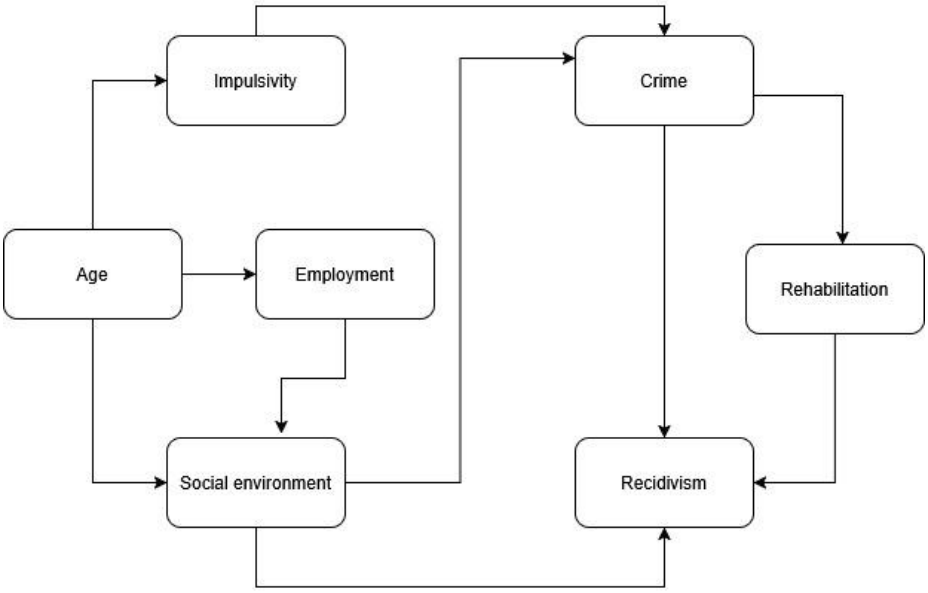
However, this relationship should be interpreted as observational rather than causal. The observed association may be influenced by unmeasured or hidden factors, meaning that age itself cannot be assumed to directly cause higher recidivism.

3. Causal Interpretation of Recidivism in the COMPAS Model

Based on the analysis of the COMPAS dataset, we have identified observational patterns such as younger individuals exhibiting higher two-year recidivism rates.

This type of result allows us to describe the behavior of the system, but it does not necessarily imply a direct causal relationship between age and recidivism. Determining which variables actually cause criminal behavior is complex, as multiple factors are involved that are not available to us or that we may not even be able to imagine.

Nevertheless, we can introduce causal reasoning, that is, propose a representation of the real world that brings us closer to those underlying mechanisms that explain these observed patterns. We can formulate the following causal model:



In this model, age does not act as a direct cause of recidivism, but rather as a factor that influences intermediate variables such as impulsivity, employment, or social environment, which in turn affect the likelihood of committing a crime and, subsequently, recidivism. Additionally, rehabilitation appears as an intervention that can reduce the probability of reoffending. Once this causal model of the world is defined, the justice system can adopt different approaches to make decisions based on this information.

3.1. Incapacitation Theory Approach

The incapacitation approach acts on the outcome (recidivism) by restricting the freedom of individuals with higher risk without intervening in the underlying causes of criminal behavior. Under this approach, the main objective of the justice system is to prevent future crimes by reducing individuals' ability to commit them, typically through incarceration.

Given that younger individuals exhibit higher recidivism rates, one possible response would be to prioritize restrictive measures for this group when the estimated risk is high. This could involve, for example, higher rates of pre-trial detention or longer sentences for young individuals deemed high-risk. However, this approach must be carefully evaluated, as it may lead to an overrepresentation of young people in prison and generate adverse long-term social effects.

3.2. Rehabilitative Theory Approach

The rehabilitative approach acts on the intermediate variables in the graph, such as impulsivity, employment, or social environment, aiming to modify the factors that cause criminal behavior. From this rehabilitative perspective, the goal is not to punish or incapacitate, but to reduce the likelihood of recidivism through targeted interventions.

Given that younger individuals show a higher probability of recidivism, an appropriate response would be to design specific programs aimed at this group, such as educational training, psychological support, job placement programs, or community-based interventions.

In this approach, age is not interpreted as a direct cause of criminal behavior, but rather as an indicator associated with factors that can be modified. Therefore, public policy would focus on reducing risk rather than restricting freedom.

3.3. Retributive Theory Approach

The retributive approach does not intervene in the causal system, but rather focuses on the committed action (crime) and the degree of individual responsibility, regardless of the factors that may have influenced it. From this theory, punishment is based on culpability, that is, on the degree of control the individual had over their actions.

The fact that younger individuals exhibit higher recidivism rates does not necessarily imply greater culpability. In fact, youth is often associated with lower maturity (impulsivity), poorer social environments, and greater external influence, which could reduce their degree of moral responsibility.

Therefore, under this approach, the justice system should not automatically increase punishment for young people solely because of their higher statistical likelihood of recidivism, but instead focus on assessing individual circumstances and intent.

3.4. Final Notes

The analysis of the relationship between age and recidivism shows how the same empirical evidence can lead to very different policies depending on the ethical framework used. While incapacitation acts on outcomes, rehabilitation intervenes on causes, and retributive justice focuses on individual responsibility.

This perspective highlights the importance of distinguishing between correlation and causation in decision-making within data-driven justice systems, as different interpretations can lead to policies with very different social impacts.